Sam Gramada

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Education

Engineering Physics, BASc (3rd Year), Minor in Commerce Aug. 2024 – May 2029
University of British Columbia *Vancouver, BC*

Technical Experience

Unmanned Mars Rover Robotics Competition-Electrical Lead June 2026 – August 2026
Technologies: KiCad, ESP32, Lidar, PID Controller, Oscilloscope, LTspice

- Secured 2nd place out of 16 teams by building an autonomous rover featuring a 4-DOF micro-stepped robotic arm, LiDAR, and Raspberry Pi object detection.
- Designed a custom ESP32 motherboard in KiCad and laid out a provided power distribution schematic, creating custom component footprints and integrating buck/boost converters for 20V/15V rails.
- Debugged and soldered custom H-bridge motor drivers on protoboards, using an oscilloscope to eliminate motor coasting and refine turn logic.
- Optimized ESP32 I/O pin allocation by networking four reflectance sensors to a shared data line, freeing up critical pins to program a custom C++ PID tape-following controller.

Agrobot Design Team - Autonomous Farming Vehicle August 2025 – September 2026
Technologies: KiCad, Solidworks, 3D Printing, Arduino, Lathe/Mill

- Built a modular weed extermination system driven by AI object detection.
- Prototyped mechanical components using SolidWorks, 3D printing, and manual machining.
- Designed KiCad PCBs to interface system electronics with an Arduino Uno.
- Fabricated a custom linear actuator with integrated position sensing.

Self Stabilizing Rotating Sonar Scanner (In Progress) September 2026
Technologies: Onshape, KiCad, LTSpice

- Designing a two-axis active stabilization platform with pitch and yaw compensation to maintain level for sensors.
- Architecting a stepper-driven rotational system to enable 360-degree continuous environmental sonar scanning.

Closed-Loop Motor Control System September 2025
Technologies: Python, JupyterLab, Oscilliscopes

- Built a 5V closed-loop control system to actively regulate electric motor speeds up to 3,690 RPM.
- Synchronized Schmitt triggers, an 8-bit counter, and a D-Latch to drive an R2R resistor ladder DAC.
- Debugged complex hardware faults and signal delays using oscilloscopes, logic analyzers, and Nodal Analysis

Other Experience

Youth Rugby Coaching | *Victoria, BC* June 2022 – June 2025

- Planned, organized, and led practices for over 50 youth athletes

Grocery Clerk | *Victoria, BC* May 2023 – May 2024

- Worked with my team to facilitate the purchasing, organize the department and help as many as 100 clients a day personally

Technical Skills

Hardware & Microcontrollers: ESP32, Raspberry Pi, Arduino, Oscilloscopes, Soldering, LiDAR, Acuators
Software & CAD: KiCad, SolidWorks, LTspice, Python, C/C+, Excel
Relevant Courses: Signals and Systems, Circuit Analysis, Linear Algebra, Control Systems
Interests: Surfing, Reading, Trail Running, Gardening